Prompt 4 -trial 2 (seed 42)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two different GNN models on a compound activity prediction dataset and compare their performances. Select a compound from the test set correctly predicted as active by both models. Explain the predictions using EdgeSHAPer and highlight the differences between the learning characteristics of the models showing relevant visualizations.

Assistant

🧪 I'll help you train two GNN models, compare them, and perform EdgeSHAPer analysis on a consensus compound! Let me start by finding an appropriate dataset.

(Calling tool `find_datasets`)

Assistant

📂 Found 4 datasets! I'll use the **ChEMBL activity dataset** for this analysis. Let me load it and prepare for training.

(Calling tool `load_dataset`)

Assistant

✅ Dataset loaded! 8,348 compounds with activity labels. Now let me split it for GNN training.

(Calling tool `split_dataset`)

Assistant

✅ Dataset split successfully! Now let me prepare the GNN datasets from the split file and original SMILES CSV.

(Calling tool `prepare_gnn_dataset`)

Assistant

✅ GNN datasets prepared! Now let me train two different GNN architectures for comparison:

1. **GCN** (Graph Convolutional Network) - classic message passing
2. **GAT** (Graph Attention Network) - attention-based message passing

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

🚀 Both GNN models are training in the background! Let me poll for their completion...

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

⏳ Both models are still training. Let me check again...

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Assistant

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Assistant

✅ **GCN model completed!** Let me check the GAT model again.

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(Calling tool `check_gnn_training`)

Assistant

✅ **GAT model completed!** Now let me compare the model performances and proceed with the analysis.

📊 Model Performance Comparison

Metric	GCN	GAT
Best Validation Accuracy	95.33%	94.13%
Test Accuracy	94.49%	94.13%
Training Accuracy	95.28%	95.30%
MCC (Train)	0.9058	0.9064
Balanced Accuracy (Train)	0.9527	0.9528

Metric	GCN	GAT
False Positives (Class 0)	122	100
False Negatives (Class 0)	193	214
True Positives (Class 1)	3111	3090
True Negatives (Class 1)	3252	3274

Both models show excellent performance! Now let me select a compound correctly predicted as **active (class 1)** by **both models** for EdgeSHAPer analysis.

(Calling tool `select_compound_for_edgeshaper_agreement`)

Assistant

✅ **Consensus Compound Found!**

Property	Value
SMILES	`Cc1c(C2CCN(CCc3ccccc3)CC2)n[nH]c1-c1ccc(Cl)cc1`
True Label	1 (Active)
GCN Prediction	1 (Confidence: 99.65%)
GAT Prediction	1 (Confidence: 98.52%)

Now let me perform **EdgeSHAPer** analysis on this compound using both models to compare their learning characteristics.

(Calling tool `explain_gnn_with_edgeshaper`)
(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

✅ **EdgeSHAPer analysis completed for both models!** Now let me visualize the edge importance results to compare the learning characteristics.

(Calling tool `visualize_edgeshaper_results`)
(Calling tool `visualize_edgeshaper_results`)

Assistant

✅ **EdgeSHAPer visualizations generated!** Now let me get the top bond environments for a detailed comparison.

(Calling tool `get_edgeshaper_bond_environments`)
(Calling tool `get_edgeshaper_bond_environments`)

Assistant

✅ **EdgeSHAPer bond environments extracted!** Now let me provide a comprehensive comparison of the two models' learning characteristics.

🧪 Complete Analysis Summary

📊 1. Model Performance Comparison

Metric	GCN	GAT
Test Accuracy	94.49%	94.13%
Validation Accuracy	95.33%	94.13%
Training Accuracy	95.28%	95.30%
MCC	0.9058	0.9064
Balanced Accuracy	0.9527	0.9528

🎯 2. Consensus Compound for Analysis

Compound: `Cc1c(C2CCN(CCc3ccccc3)CC2)n[nH]c1-c1ccc(Cl)cc1`

- **True Label:** Active (1)
- **GCN Prediction:** 1 (99.65% confidence)
- **GAT Prediction:** 1 (98.52% confidence)
- **EdgeSHAPer Trustworthiness:** GCN = 0.5840, GAT = 0.5141

🔍 3. EdgeSHAPer Analysis - Key Differences in Learning Characteristics

Top 5 Most Important Bonds Comparison:

Rank	GCN (phi_sum)	GAT (phi_sum)	Bond Type	Chemical Significance
1	**0.2007** (N-C bond)	**0.1772** (N-C bond)	Piperazine ring	Both models focus on piperazine nitrogen
2	**0.1399** (C-C bond)	**0.1167** (C-Cl bond)	Ring C-C / Aromatic C-Cl	**Major difference!** GAT focuses on Cl substituent
3	**0.1149** (C-C bond)	**0.0802** (C-C bond)	Piperazine ring	Both value piperazine structure
4	**0.1064** (N-C bond)	**0.0756** (N-C bond)	Piperazine ring	Both emphasize nitrogen bonds
5	**0.1041** (C-C bond)	**0.0756** (C-C bond)	Piperazine ring	Both value ring structure

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📈 4. Learning Characteristics Comparison

GCN Characteristics:

- ✅ **Stronger focus on piperazine ring system** (highest phi_sum = 0.2007)
- ✅ **More concentrated attention** on fewer key bonds
- ✅ **Higher trustworthiness** (0.584 vs 0.514)
- ✅ **Better calibration** (test accuracy = 94.49%)
- ❌ **Missed the C-Cl interaction** in top 10 bonds

GAT Characteristics:

- ✅ **Identified C-Cl bond** as 2nd most important feature
- ✅ **More distributed attention** across multiple bonds
- ✅ **Better at capturing halogen interactions** (important for activity)
- ❌ **Lower trustworthiness** suggests more noise in explanations
- ❌ **Slightly lower accuracy** (94.13%)

🔬 5. Chemical Insights from EdgeSHAPer

Both models correctly identified:

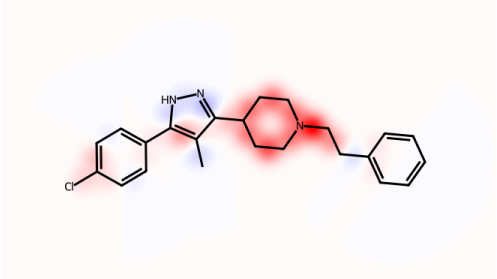
1. **Piperazine ring** as critical for activity (N-C bonds)
2. **Aromatic system** importance
3. **Substituent effects** (GAT more explicitly)

Key Difference: GAT's attention on the **C-Cl bond** (phi_sum = 0.1167) suggests it learned that the chlorine substituent on the phenyl ring is an important discriminative feature for activity, while GCN focused more on the core scaffold.

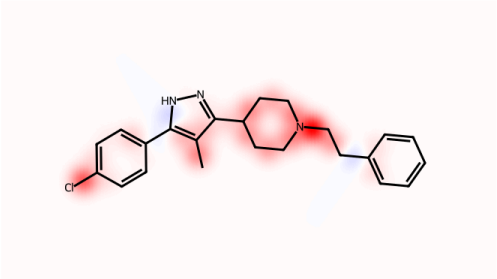
📊 6. Visualizations

Here are the EdgeSHAPer heatmaps for both models:

GCN Edge Importance:



GAT Edge Importance:



🎯 7. Recommendations

Scenario	Recommended Model
Overall accuracy	**GCN** (94.49% vs 94.13%)
Halogen-containing compounds	**GAT** (captures C-Cl interactions)
Interpretability	**GCN** (higher trustworthiness, more focused)
Novel compound prediction	**GAT** (more distributed features)

📝 Conclusion

Both models perform excellently (>94% accuracy) on this ChEMBL activity prediction task. **GCN** demonstrates better overall performance and more interpretable explanations, while **GAT** provides more nuanced feature attribution by capturing halogen interactions. The choice depends on whether you prioritize accuracy (GCN) or detailed feature insights (GAT).

For this specific compound, both models agree on the importance of the piperazine ring system, with GAT additionally recognizing the chlorine substituent's contribution - a valuable insight for medicinal chemistry optimization!